

Random Variable

A Random Variable is a function from Sample Space (S) to set of all real no/s R .

i.e., $X: S \rightarrow R$ Where S — Domain Space
 R — Range Space

* Types of Random Variable

* Discrete Random Variable

* Continuous Random Variable.

Discrete Random Variable

If X is a R.V which can take a finite number (or) countably infinite number of values then X is called as a discrete R.V.

when X is a discrete R.V then the possible values of X may be assumed as $x_1, x_2, \dots, x_n, \dots$

Example

The no. shown when a die is thrown and the no. of alpha particles emitted by a radioactive source are discrete R.V.

Continuous R.V.

A random variable X is said to be continuous if it can take all possible values (in an interval) between certain limits.

Example

- * Age of a person, height & weight of the set of persons.
- * life time of a bulb. etc.

Note

- * If x_1 & x_2 are R.V's & c is a Constant then cx_1 , $x_1 + x_2$, $x_1 x_2$ & $c_1 x_1 + c_2 x_2$ are also random variable where c_1 & c_2 are constants.
- * $x_1 - x_2$ is also a r.v
- * If x is a r.v then
 - $1/x$ where $1/x = \infty$ if $x = 0$
 - $x_+ = \max\{0, x\}$
 - $x_- = \min\{0, x\}$
- & iv) $|x|$ are also R.V's.
- * If x_1 & x_2 are RV's then
 - $\max[x_1, x_2]$
 - $\min[x_1, x_2]$are also Random variables.

(12)

* If x is r.v & f is a continuous (or) increasing function then $f(x)$ is also a r.v.

Formula

Discrete R.v

① Probability mass function

$$i) P(x) = \begin{cases} P(x=x_i) = p_i & \text{if } x=x_i \\ 0 & \text{if } x \neq x_i \end{cases}$$

$$ii) p_i \geq 0 \quad \forall i$$

$$iii) \sum p_i = 1$$

iv)

$x=x_i$	$P(x=x_i)$
x_1	p_1
x_2	p_2
$\vdots$	$\vdots$
x_r	p_r
$\vdots$	$\vdots$

Continuous R.v

Probability density function

$$i) P(x < x \leq x+dx) = f(x)dx$$

$$ii) f(x) \geq 0$$

$$iii) \int_R f(x) dx = 1$$

$$iv) P(a \leq x \leq b) = P(a < x < b) = \int_a^b f(x) dx$$

$$v) P(a \leq x \leq b) = P(a < x \leq b) = P(a \leq x < b) = P(a < x < b)$$

② Cumulative Distribution function (CDF)

For discrete R.v

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} p_i$$

For Continuous R.v

$$F(x) = P(-\infty \leq X \leq x) = \int_{-\infty}^x f(x) dx$$

Discrete continuous

$$(3) \quad P(x_i) = F(x_i) - F(x_{i-1}) \quad \cdot \quad \frac{d}{dn} F(n) = f(n) = F'(n)$$

$$(4) \quad \text{Mean} = E(x) = \sum_i x_i p_i$$

$$E(x) = \int_{-\infty}^{\infty} x f(n) dn$$

$$E(x^2) = \sum_i x_i^2 p_i$$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(n) dn$$

$$\text{Var}(x) = E(x^2) - (E(x))^2 = M_2$$

$$V(x) = E(x^2) - (E(x))^2 = M_2$$

Properties of cdf $F(n)$

$$(1) \quad 0 \leq F(x) \leq 1$$

$$(2) \quad F(x) \leq F(y) \quad \text{if } x < y$$

i.e., $F(n)$ is monotonically non-decreasing function & lie between 0 & 1.

$$(3) \quad F(-\infty) = 0 \quad \& \quad F(\infty) = 1.$$

Various Measures of Central tendency, Dispersion, Skewness & kurtosis for Continuous Probability Distribution & Discrete Probability Distribution

Discrete

$$\textcircled{1} \text{ AM} = E(x) = \sum x_i p_i$$

$$\textcircled{2} \text{ HM}$$

$$\text{HM} = \sum \frac{1}{x_i} p_i$$

$$\textcircled{3} \text{ G.M}$$

$$\log G = \sum \log x_i p_i$$

$$\textcircled{4} \text{ Moment about origin}$$

$$M_r' = \sum x_i^r p_i$$

= r^{th} moment about origin

Continuous

$$\text{AM} = \int_{-\infty}^{\infty} x f(x) dx$$

If Interval is $[a, b]$

$$\text{AM} = \int_a^b x f(x) dx$$

$$\text{HM} = \int_a^b \frac{1}{x} f(x) dx$$

$$\log G = \int_a^b \log x f(x) dx$$

$$M_r' = r^{\text{th}} \text{ moment about origin}$$

$$= \int_a^b x^r f(x) dx$$

r^{th} moment about the point A

$$= M_r' = \sum_i (x_i - A)^r p_i$$

$$M_r' = \int_a^b (x - A)^r f(x) dx$$

r^{th} moment about the mean

$$M_r = \sum_i (x_i - \bar{x})^r p_i$$

$$M_r = \int_a^b (x - \bar{x})^r f(x) dx$$

For $r=1$ & 2

$$M_1' = \sum_i x_i p_i$$

$$M_1' = \int_a^b x f(x) dx$$

$$M_2' = \sum_i x_i^2 p_i$$

$$M_2' = \int_a^b x^2 f(x) dx$$

$$\text{Var}(x) = M_2 = M_2' - (M_1')^2$$

$$V(x) = M_2 = M_2' - (M_1')^2$$

Note (For continuous case)

① Median

$$\int_a^M f(x) dx = \int_M^b f(x) dx = 1/2$$

$\therefore$ By solving $\int_a^M f(x) dx = 1/2$ (or) $\int_M^b f(x) dx = 1/2$ for M , we get the value of the median.

* Mean deviation about mean

$$MD = \int_a^b |x - \text{mean}| f(x) dx$$

* Mean deviation about an Average 'A'

$$MD \text{ about } A' = \int_a^b |x - A'| f(x) dx$$

* Quartiles & Deciles

Q_1 & Q_3 are given by the equations

$$\int_a^{Q_1} f(x) dx = 1/4 \quad \& \quad \int_a^{Q_3} f(x) dx = 3/4$$

* D_i , i^{th} decile is given by,

$$\int_a^{D_i} f(x) dx = i/10, \quad i = 1, 2, \dots, 9$$

* Mode: Mode is the value of x for which $f(x)$ is maximum. Mode is thus the solution of $f'(x) = 0$ & $f''(x) < 0$ provided it lies in $[a, b]$.

Note

* For discrete case, to find median, mode, M.D, Quartiles & Deciles, Mode, replace $\int$ as $\sum$ & $f(x)$ as $p(x)$.

Problems

① A Random Variable x has the following probability function

$x:$ 0 1 2 3 4 5 6 7

$P(x):$ 0 k $2k$ $2k$ $3k$ k^2 $2k^2$ $7k^2+k$

then find i) $k=?$ ii) Evaluate $P(X < 6)$,

$P(X \geq 6)$ & $P(0 < X < 5)$, iii) $P(X \leq a) > \frac{1}{2}$ find the minimum value of a ,

iv) Determine the distribution function of x

v) Find the mean & Variance.

Sol

i) Since $\sum p_i = 1$

$$k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k - 1 = 0$$

$$10(k - \frac{1}{10})(k + 1) = 0$$

$$\therefore k = \frac{1}{10} \text{ (or) } -1 \text{ (Rejected)}$$

$$\begin{array}{c} \nearrow 10 \\ -1x \quad 10 \\ 10 \quad 10 \\ \downarrow \\ -\frac{1}{10} \times 1 \end{array}$$

$$\therefore k = 1/10$$

iv) Distribution table with distribution function

x	0	1	2	3	4	5	6	7
$P(x)$	0	$1/10$	$2/10$	$2/10$	$3/10$	$1/100$	$2/100$	$17/100$
$F(x) = P(X \leq x)$	0	$1/10$	$3/10$	$5/10$	$8/10$	$81/100$	$83/100$	1

$$\begin{aligned} \textcircled{ii} \quad P(X < 6) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &\quad + P(X=4) + P(X=5) \\ &= \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100} = \frac{81}{100} \end{aligned}$$

$$P(X \geq 6) = 1 - P(X < 6) = 1 - \frac{81}{100} = \frac{19}{100}$$

$$\begin{aligned} P(0 < X < 5) &= P(X=1) + P(X=2) + P(X=3) + P(X=4) \\ &= \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} \\ &= \frac{8}{10} = \frac{4}{5} \end{aligned}$$

iii) $P(X \leq a) > \frac{1}{2}$ when a is minimum

$$P(X \leq 3) = \frac{5}{10} = 0.5 = \frac{1}{2}$$

$$P(X \leq 4) = \frac{8}{10} = 0.8 > \frac{1}{2}$$

$$\therefore a = 4$$

$$v) \text{ Mean} = E(x) = \sum x_i p_i$$

$$= \frac{1}{10} + \frac{4}{10} + \frac{6}{10} + \frac{12}{10} + \frac{5}{100} + \frac{12}{100} + \frac{119}{100}$$

$$= \frac{23}{10} + \frac{136}{100} = \frac{230+136}{100}$$

$$= \frac{366}{100}$$

$$E(x^2) = \frac{1}{10} + \frac{8}{10} + \frac{18}{10} + \frac{48}{10} + \frac{25}{100} + \frac{72}{100} + \frac{833}{100}$$

$$= \sum x_i^2 p_i = \frac{75}{10} + \frac{930}{100}$$

$$= \frac{750+930}{100} = \frac{1680}{100}$$

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

$$= \frac{1680}{100} - \left(\frac{366}{100}\right)^2$$

$$= 16.80 - 13.40 = 3.4$$

Problem - 2

$$\text{If } p(x) = \begin{cases} x/15 & : x = 1/2, 3, 4, 5 \\ 0 & \text{elsewhere} \end{cases}$$

then find i) $P\{x=1 \text{ (or) } 2\}$

ii) $P\{1/2 < x < 5/2 \mid x > 1\}$

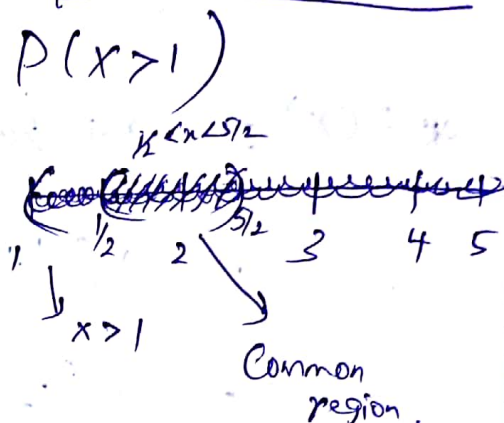
Sol

$$i) P(X=1 \text{ (or) } 2) = P(X=1) + P(X=2) \\ = \frac{1}{15} + \frac{2}{15} = \frac{1}{5}$$

$$ii) P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) = \frac{P\left(\left(\frac{1}{2} < X < \frac{5}{2}\right) \cap (X > 1)\right)}{P(X > 1)}$$

$$= \frac{P\left((X=1 \text{ (or) } 2) \cap X > 1\right)}{P(X > 1)} \\ = \frac{P\left(\frac{1}{2} < X < \frac{5}{2}\right)}{P(X > 1)}$$

$$= \frac{P(X=2)}{1 - P(X=1)} = \frac{\frac{2}{15}}{1 - \frac{1}{15}} = \frac{1}{7}$$



② The probability function of an infinite discrete distribution is given by $P(X=j) = \frac{1}{2^j}$ ($j=1, 2, \dots, \infty$). Verify that the total probability is 1 & find the mean and variance of the distribution. Find also $P(X \text{ is even})$, $P(X \geq 5)$ & $P(X \text{ is divisible by } 3)$.

Sol

$$P(X=j) = p_j = \frac{1}{2^j}$$

i) Verification of $\sum p_j = 1$.

$$\sum p_i = \frac{1}{2} + \frac{1}{2}^2 + \dots \infty \quad \left(\text{Which is a GP with common ratio } r = \frac{1}{2} \right)$$

$$S_{\infty} = \frac{a}{1-r} \quad (\because a = \frac{1}{2}, r = \frac{1}{2})$$

$$= \frac{\frac{1}{2}}{1 - \frac{1}{2}}$$

$$= 1$$

$$* E(X) = a + 2a^2 + 3a^3 + \dots \infty \quad (\text{where } a = \frac{1}{2})$$

$$= a(1 + 2a + 3a^2 + \dots \infty)$$

$$= a(1-a)^{-2} = \frac{a}{(1-a)^2} = \frac{\frac{1}{2}}{(1-\frac{1}{2})^2} = 2$$

$$* E(X^2) = \sum_j j^2 p_j$$

$$= \sum_j [j(j+1) - j] a^j$$

$$= \sum_{j=1}^{\infty} j(j+1) a^j - \sum_{j=1}^{\infty} j a^j$$

$$= a(1 \cdot 2 + 2 \cdot 3a + 3 \cdot 4a^2 + \dots \infty) - a(1 + 2a + 3a^2 + \dots \infty)$$

$$= \frac{a \cdot 2}{(1-a)^3} - \frac{a}{(1-a)^2} = 8 - 2 = 6$$

$$V(X) = E(X^2) - (E(X))^2 = 6 - 4 = 2$$

$$* P(X \text{ is even}) = P(X=2 \text{ (or) } X=4 \text{ (or) } X=6 \text{ (or) etc})$$

$$= P(X=2) + P(X=4) + \dots + \infty$$

$$= \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^6 + \dots + \infty$$

$$= \frac{\left(\frac{1}{4}\right)}{\left(1 - \frac{1}{4}\right)} = \frac{1}{3}$$

$$* P(X \geq 5) = P(X=5 \text{ (or) } X=6 \text{ (or) } X=7 \text{ (or) etc})$$

$$= P(X=5) + P(X=6) + \dots + \infty$$

$$= \frac{\frac{1}{2^5}}{1 - \frac{1}{2}} = \frac{1}{16}$$

$$* P(X \text{ is divisible by } 3)$$

$$= P(X=3 \text{ (or) } X=6 \text{ (or) } X=9 \text{ etc})$$

$$= P(X=3) + P(X=6) + \dots$$

$$= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^9 + \dots$$

$$= \frac{\frac{1}{8}}{1 - \frac{1}{8}} = \frac{1}{7}$$

④ If the density function of a continuous R.V x is given by,

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ 3a - ax, & 2 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

i) Find the value of a

ii) Find the Cdf of x

~~Sol~~
Sol

$$\int_0^3 f(x) dx = 1$$

$$\int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (3a - ax) dx = 1$$

$$a \left(\frac{x^2}{2} \right)_0^1 + a(x)_1^2 + \left(3ax - a \frac{x^2}{2} \right)_2^3 = 1$$

$$a \left(\frac{1}{2} - 0 \right) + a(2-1) + \left[\left(9a - 9a \frac{1}{2} \right) - (6a - 2a) \right] = 1$$

$$a \left[\frac{1}{2} + 1 + \frac{9}{2} - 4 \right] = 1$$

$$a(6-4) = 1$$
$$2a = 1$$

$$\boxed{a = \frac{1}{2}}$$

ii) $F(x) = P(X \leq x) = 0$ when $x < 0$.

* $F(x) = \int_0^x x/2 \, dx = x^2/4$, when $0 \leq x \leq 1$

* In $1 \leq x \leq 2$

$$F(x) = \int_0^1 x/2 \, dx + \int_1^x 1/2 \, dx = \frac{1}{2} \left(\frac{x^2}{2} \right)_0^1 + \frac{1}{2} (x)_1^x$$

$$= \frac{1}{2} \left(\frac{1}{2} - 0 \right) + \frac{1}{2} (x - 1)$$

$$= \frac{1}{4} + \frac{x}{2} - \frac{1}{2} = \frac{x}{2} - \frac{1}{4} \quad \text{when } 1 \leq x \leq 2$$

* In $2 \leq x \leq 3$

$$F(x) = \int_0^1 x/2 \, dx + \int_1^2 1/2 \, dx + \int_2^x (3/2 - x/2) \, dx$$

$$= \frac{1}{2} \left(\frac{x^2}{2} \right)_0^1 + \frac{1}{2} (x)_1^2 + \frac{1}{2} \left(3x - \frac{x^2}{2} \right)_2^x$$

$$= \frac{1}{2} \left(\frac{1}{2} \right) + \frac{1}{2} (2 - 1) + \frac{1}{2} \left[\left(3x - \frac{x^2}{2} \right) - \left(6 - 2 \right) \right]$$

$$= \frac{1}{4} + \frac{1}{2} + \frac{1}{2} \left[3x - \frac{x^2}{2} \right] - 2$$

$$= \frac{3}{4} + \frac{1}{2} \left(3x - \frac{x^2}{2} \right) - 2$$

$$= \frac{3}{2}x - \frac{x^2}{4} - \frac{5}{2} \quad \text{When } 2 \leq x \leq 3$$

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$$F(x) = 1 \quad \text{when } x > 3.$$

⑤ The diameter of an electric cable say x is assumed to be a Continuous random variable with Pdf $f(x) = kx(1-x)$, $0 \leq x \leq 1$

then i) find k ii) Determine a number

'b' such that $P(x < b) = P(x > b)$

iii) find mean & var(x)

Sol

$$i) \int_0^1 f(x) dx = 1$$

$$\therefore k \int_0^1 (x - x^2) dx = 1$$

$$\therefore k \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^1 = 1$$

$$k \left(\frac{1}{2} - \frac{1}{3} \right) = 1$$

$$k \left(\frac{1}{6} \right) = 1$$

$$\therefore k = 6$$

$$\therefore f(x) = 6x(1-x), \quad 0 \leq x \leq 1$$

$$ii) P(x < b) = P(x > b)$$

$$\int_0^b f(x) dx = \int_b^1 f(x) dx$$

$$\left(\int_0^b (x - x^2) dx \right) = \int_b^1 (x - x^2) dx$$

$$\left(x^2/2 - x^3/3 \right)_0^b = \left(x^2/2 - x^3/3 \right)_b^1$$

$$b^2/2 - b^3/3 = \left(1/2 - 1/3 \right) - \left(b^2/2 - b^3/3 \right)$$

$$2b^2/2 - 2b^3/3 = 1/6$$

$$\frac{3b^2 - 2b^3}{3} = 1/6$$

$$3b^2 - 2b^3 = 1/2$$

$$3b^2 - 2b^3 - 1/2 = 0$$

$$6b^2 - 4b^3 - 1 = 0$$

$$\text{i.e., } 4b^3 - 6b^2 + 1 = 0$$

By trial & error, $b = 1/2$ is a root

Since

$$\frac{1}{2} \left| \begin{array}{ccc|c} 4 & -6 & 0 & 1 \\ 0 & 2 & -2 & -1 \\ \hline 4 & -4 & -1 & 0 \end{array} \right|$$

$$4b^3 - 6b^2 + 1 = 0$$

$$(b - 1/2)(4b^2 - 4b - 1) = 0$$

$$\therefore \boxed{b = 1/2}$$

Mathematical Expectation

Discrete R.V

$$E(x) = \text{Expected value of } x \\ = \sum_{i=1}^n x_i p_i$$

Continuous R.V

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

Two dimensional Random Variables

Let X & Y be two Random Variables defined on the same sample space S then the function (X, Y) that assigns a point in $R^2 (= R \times R)$ is called a two dimensional Random Variable.

Note

Let $A = \{a < x \leq b\}$ & $B = \{c < y \leq d\}$ be two events then

$$\{a < x \leq b, c < y \leq d\} = \{a < x \leq b\} \cap \{c < y \leq d\} \\ = A \cap B$$

$$\therefore P(\{a < x \leq b, c < y \leq d\}) = P(A \cap B)$$

Discrete case

① Joint Probability mass function

$$P_{xy}(x_i, y_j) \text{ (or) } P(x_i, y_j) \\ = P(X=x_i, Y=y_j)$$

② Marginal Probability function

$$P(X=x_i) = P_{i.} \\ = \sum_{j=1}^m p_{ij} = p_{i1} + p_{i2} + \dots + p_{im} \\ = \text{Marginal probability distribution of } X.$$

Similarly Marginal probability distribution of Y

$$P(Y=y_j) = P_{.j} \\ = \sum_{i=1}^n p_{ij} = p_{1j} + p_{2j} + \dots + p_{nj}$$

$$\text{(or)} \quad \sum_y P(x_i, y) = P(X=x_i) \\ \sum_x P(x, y_j) = P(Y=y_j)$$

Continuous case

Joint pdf

$$f(x, y)$$

Marginal density function

Marginal density function of X

$$= f(x) \\ = \int_{-\infty}^{\infty} f(x, y) dy$$

Marginal density function of Y

$$= f(y) \\ = \int_{-\infty}^{\infty} f(x, y) dx$$

Note

$$f(x) = \frac{dF_x(x)}{dx} \\ f(y) = \frac{dF_y(y)}{dy}$$

Note

Cumulative distribution function for 2D R.V.

$$F(x, y) = P(X \leq x, Y \leq y) \text{ \& } f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

Discrete

Continuous

i) Marginal distribution function of X & Y respectively in 2D R.V.

$$F(x) = \sum_y P(X \leq x, Y = y)$$

$$\& F(y) = \sum_x P(X = x, Y \leq y)$$

$$i) F(x) = \int_{-\infty}^x \int_{-\infty}^{\infty} f(x, y) dy dx$$

$$F(y) = \int_{-\infty}^y \int_{-\infty}^{\infty} f(x, y) dx dy$$

ii) Cumulative distributive function

$$F(x, y) = \sum_{y_j \leq y} \sum_{x_i \leq x} p_{ij}$$

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(x, y) dx dy$$

iii) Total Probability

$$\sum_i \sum_j p_{ij} = 1$$

provided $p_{ij} \geq 0$

for all i & j

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

provided

$f(x, y) \geq 0$
for all $x, y \in \mathbb{R}$.

Discrete

iv) Conditional probability Distribution

$$\begin{aligned} P(X=x_i | Y=y_j) \\ &= \frac{P(X=x_i, Y=y_j)}{P(Y=y_j)} \\ &= \frac{p_{ij}}{p_{*j}} \end{aligned}$$

$$\begin{aligned} \text{III}^{\text{ly}} P(Y=y_j | X=x_i) \\ &= \frac{P(X=x_i, Y=y_j)}{P(X=x_i)} \\ &= \frac{p_{ij}}{p_{i*}} \end{aligned}$$

are the Conditional
probability distributions
of X given Y &
 Y given X respectively.

(2) Continuous

$$f(x/y) = \frac{f(x, y)}{f(y)}$$

$$\& f(y/x) = \frac{f(x, y)}{f(x)}$$

are the Conditional
density functions of
 x given y & y given
 x respectively.

Discrete

Condition for Independent

$$\text{If } p_{ij} = p_{i\cdot} \times p_{\cdot j} \quad \forall i, j$$

then X & Y are said to be independent Random Variables.

Continuous

If $f(x, y) = f(x) \cdot f(y)$
then X & Y are said to be independent R.V.'s

Expected Values of a Two-Dimensional Random Variable. (X, Y)

Discrete

$$E(XY) = \sum \sum x_i y_j p_{ij}$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E(Y) = \int_{-\infty}^{\infty} y f(y) dy$$

(for continuous)

Properties

- i) $E(X+Y) = E(X) + E(Y)$
- ii) $E(XY) \neq E(X) \cdot E(Y)$ in general.

Continuous

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy$$

For discrete

$$E(X) = \sum x p(x)$$

$$E(Y) = \sum y p(y)$$

iii) If x & y are independent R.V's then
 $E(XY) = E(X) \cdot E(Y)$

Conditional Expected value

Discrete

$$E(g(x,y) | y=y_j) \\ = \sum_i g(x_i, y_j) \cdot P(X=x_i | Y=y_j)$$

$$||^{ly} E(g(x,y) | x=x_i) \\ = \sum_j g(x_i, y_j) \cdot P(Y=y_j | X=x_i)$$

Continuous

$$E(g(x,y) | y) \\ = \int_{-\infty}^{\infty} g(x,y) \cdot f(x/y) dx$$

$$E(g(x,y) | x) \\ = \int_{-\infty}^{\infty} g(x,y) \cdot f(y/x) dy$$

Sums

- ① The Joint Probability mass function of (X, Y) is given by $P(X, Y) = k(2x + 3y)$, $x = 0, 1, 2$, $y = 1, 2, 3$. Find all the marginal & Conditional & probability distributions. Also i) find the probability distribution of $x+y$, ii) verify whether x & y are independent.
- iii) Find Mean of X & Mean of Y , $Var(X)$ & $Var(Y)$
- iv) $cov(X \& Y)$, v) coefficient of correlation between x & y .

Note

$$\rho = \text{coefficient of correlation} \\ = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Sol

X \ Y	1	2	3
0	3k	6k	9k
1	5k	8k	11k
2	7k	10k	13k

To find: $k = ?$

$$\sum_i \sum_j P_{ij} = 1$$

$$72k = 1$$

$$k = 1/72$$

Marginal Probability distribution of X

$$i) P(X=0) = P_{0*} = P_{01} + P_{02} + P_{03}$$

$$= P(X=0, Y=1) + P(X=0, Y=2) + P(X=0, Y=3)$$

$$= 18k = 18/72$$

$$\begin{aligned} \text{ii) } P(X=1) &= P_{1k} = P_{11} + P_{12} + P_{13} \\ &= P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) \\ &= 24/72 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(X=2) &= P_{2k} = P_{21} + P_{22} + P_{23} \\ &= 30/72 \end{aligned}$$

Marginal probability distribution of Y:

$$\text{i) } P(Y=1) = P_{yk} = 15/72$$

$$\text{ii) } P(Y=2) = 24/72$$

$$\text{iii) } P(Y=3) = 33/72$$

i.e.,

X \ Y	Y			Marginal distribution of X
	1	2	3	
0	3/72	6/72	9/72	$P(X=0) = 18/72$ $P(X=1) = 24/72$ $P(X=2) = 30/72$
1	5/72	8/72	11/72	
2	7/72	10/72	13/72	
Marginal distribution of Y	$P(Y=1) = 15/72$	$P(Y=2) = 24/72$	$P(Y=3) = 33/72$	1 → Total Probability

Conditional Probability Distribution:

i) Conditional Probability of x given y=1:

$$P(X=0|Y=1) = \frac{P(X=0, Y=1)}{P(Y=1)} = \frac{P_{01}}{P(Y=1)}$$

$$= \frac{3/72}{15/72} = \frac{3}{15} = \frac{1}{5}$$

$$P(X=1|Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{P_{11}}{P(Y=1)} = \frac{5/72}{15/72} = \frac{1}{3}$$

$$P(X=2|Y=1) = \frac{7}{15}$$

ii) Conditional Probability of X given Y=2:

$$P(X=0|Y=2) = \frac{P(X=0, Y=2)}{P(Y=2)} = \frac{1}{4}$$

$$P(X=1|Y=2) = \frac{P(X=1, Y=2)}{P(Y=2)} = \frac{8k}{24k} = \frac{1}{3}$$

$$P(X=1|Y=3) = \frac{P(X=1, Y=3)}{P(Y=3)} = \frac{P_{13}}{P(Y=3)} = \frac{10k}{24k} = \frac{5}{12}$$

iii) Conditional distribution of X given Y=3:

$$P(X=0 | Y=3) = \frac{9k}{33k} = \frac{3}{11}$$

$$P(X=1 | Y=3) = \frac{11k}{33k} = \frac{1}{3}$$

$$P(X=2 | Y=3) = \frac{13k}{33k} = \frac{13}{33}$$

iv) Conditional distribution of Y given X=0:

$$P(Y=1 | X=0) = \frac{P(X=0, Y=1)}{P(X=0)} = \frac{3k}{18k} = \frac{1}{6}$$

$$P(Y=2 | X=0) = \frac{6k}{18k} = \frac{1}{3}$$

$$P(Y=3 | X=0) = \frac{9k}{18k} = \frac{1}{2}$$

v) Conditional distribution of Y given X=1:

$$P(Y=1 | X=1) = \frac{P(X=1, Y=1)}{P(X=1)} = \frac{5k}{24k} = \frac{5}{24}$$

$$P(Y=2 | X=1) = \frac{8k}{24k} = \frac{1}{3}$$

$$P(Y=3 | X=1) = \frac{11k}{24k} = \frac{11}{24}$$

ii) Conditional distribution of y given $x=2$:

$$P(Y=1 | X=2) = \frac{7k}{30k} = \frac{7}{30}$$

$$P(Y=2 | X=2) = \frac{10k}{30k} = \frac{1}{3}$$

$$P(Y=3 | X=2) = \frac{13k}{30k} = \frac{13}{30}$$

Probability distribution of $(x+y)$

$x+y$	$P(x+y)$
1	$P_{01} = \frac{3}{72}$
2	$P_{02} + P_{11} = \frac{11}{72}$
3	$P_{03} + P_{12} + P_{21} = \frac{24}{72}$
4	$P_{13} + P_{22} = \frac{21}{72}$
5	$P_{23} = \frac{13}{72}$

ii) Verification for X & Y are independent
(or) not !

Condition : $P_{ij} = P_{i*} \times P_{*j}$ for all i, j

$$P_{11} = 5/72 \quad \left| \quad \begin{aligned} P_{1*} &= P(X=1) = 11/72 \\ P_{*1} &= P(Y=1) = 15/72 \end{aligned} \right.$$

$$P_{11} \neq P_{1*} \times P_{*1}$$

$\therefore X$ & Y are not independent.

iii) Mean of X & Mean of Y

$$E(X) = \sum_{i=0}^2 x_i p_i = 0 \times 18/72 + 1 \times 24/72 + 2 \times 30/72$$

$$= \frac{24}{72} + \frac{60}{72} = \frac{84}{72}$$

$$E(Y) = \sum_{j=1}^3 y_j p_j = 1 \times P(Y=1) + 2 \times P(Y=2) + 3 \times P(Y=3)$$

$$= 1 \times 15/72 + 2 \times 24/72 + 3 \times 33/72$$

$$= 15/72 + 48/72 + 99/72 = \frac{162}{72}$$

$$iv) \text{Cov}(x, y) = E(xy) - E(x) \cdot E(y)$$

$$E(xy) = \sum_i \sum_j x_i y_j p_{ij}$$

$$= 0 \times 1 \times \frac{3}{72} + 0 \times 2 \times \frac{6}{72} + 0 \times 3 \times \frac{9}{72} \\ + 1 \times 1 \times \frac{5}{72} + 1 \times 2 \times \frac{8}{72} + 1 \times 3 \times \frac{11}{72} \\ + 2 \times 1 \times \frac{7}{72} + 2 \times 2 \times \frac{10}{72} + 2 \times 3 \times \frac{13}{72}$$

$$= 0 + 0 + 0 + \frac{5}{72} + \frac{16}{72} + \frac{33}{72} \\ + \frac{14}{72} + \frac{40}{72} + \frac{78}{72}$$

$$= \frac{206}{72}$$

$$v) \rho(x, y) = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\sigma_x = \sqrt{V(x)} \quad \text{where } V(x) = E(x^2) - (E(x))^2$$

$$\sigma_y = \sqrt{V(y)} \quad \text{where } V(y) = E(y^2) - (E(y))^2$$

— Do as Exercise.

(26)

② The joint pdf of a 2D R.V. (x, y) is

given by
$$f(x, y) = \begin{cases} xy^2 + x^2/8, & 0 \leq x \leq 2, \\ & 0 \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

then compute i) $P(x > 1)$, $P(y < 1/2)$,

$P(x > 1 \mid y < 1/2)$, $P(y < 1/2 \mid x > 1)$ &

ii) $P(x + y \leq 1)$

Sol

i) $P(x > 1) = \int_{y=0}^1 \int_{x=1}^2 f(x, y) dx dy$

$$= \int_0^1 \int_1^2 (xy^2 + x^2/8) dx dy = \int_0^1 \left[\frac{x^2}{2} y^2 + \frac{x^3}{24} \right]_1^2 dy$$

$$= \int_0^1 \left[\left(2y^2 + \frac{1}{3} \right) - \left(\frac{y^2}{2} + \frac{1}{24} \right) \right] dy$$

$$= \int_0^1 \left(2y^2 + \frac{8}{24} - \frac{y^2}{2} - \frac{1}{24} \right) dy = \int_0^1 \left(\frac{3y^2}{2} + \frac{7}{24} \right) dy$$

$$= \left[\frac{2}{2} y^3 + \frac{7}{24} y^4 \right]_0^1 = \frac{1}{2} + \frac{7}{24}$$

$$= \frac{12+7}{24} = \frac{19}{24}$$

$$P(Y < 1/2) = \int_{y=0}^{1/2} \int_{x=0}^2 (xy^2 + x^2/8) dx dy$$

$$= 1/4$$

$$P(X > 1, Y < 1/2) = \int_{y=0}^{1/2} \int_{x=1}^2 (xy^2 + x^2/8) dx dy$$

$$= 5/24$$

$$P(X > 1 | Y < 1/2) = \frac{P(X > 1, Y < 1/2)}{P(Y < 1/2)}$$

$$= \frac{5/24}{1/4} = 5/6$$

$$P(Y < Y_2 | X > 1) = \frac{P(X > 1, Y < Y_2)}{P(X > 1)}$$

$$= \frac{5/24}{19/24} = 5/19$$

$$P(X < Y) = \int_0^1 \int_0^y \left(xy^2 + \frac{x^2}{8} \right) dx dy$$

$$= 53/480$$

$$\text{ii) } P(X+Y < 1) = \int_0^1 \int_0^{1-y} \left(xy^2 + \frac{x^2}{8} \right) dx dy \quad \left(\begin{array}{l} \because X+Y=1 \\ X=1-Y \end{array} \right)$$

$$= 13/480$$

③ The joint pdf of (x, y) is given by $f(x, y) = kxy e^{-(x^2+y^2)}$, $x > 0$, $y > 0$. Find the value of k & verify whether x & y are independent.

Sol i) $\int_0^\infty \int_0^\infty f(x, y) dx dy = 1$

$$k \int_0^\infty x e^{-x^2} dx \cdot \int_0^\infty y e^{-y^2} dy = 1$$

$$k \cdot 1/2 \cdot 1/2 = 1$$

$$k/4 = 1$$

$$k = 4$$

ii) $f(x) = \int_{y=0}^\infty f(x, y) dy = 4 \int_0^\infty x y e^{-x^2} \cdot e^{-y^2} dy$

$$= 4 x e^{-x^2} \int_0^\infty y e^{-y^2} dy = 4 x e^{-x^2} \cdot 1/2 = 2 x e^{-x^2}$$

iii) $f(y) = 2 y e^{-y^2}$

$$\therefore f(x, y) = 4 x y e^{-x^2} \cdot e^{-y^2} \quad \text{--- (1)}$$

$$f(x) \cdot f(y) = 2 x e^{-x^2} \cdot 2 y e^{-y^2}$$

$$= 4 x y e^{-x^2} \cdot e^{-y^2} \quad \text{--- (2)}$$

from (1) & (2) x & y are independent.

$$t = x^2$$

$$dt = 2x dx$$

$$dt/2 = x dx$$

$x:$	0	∞
$t:$	0	∞

$$\int_0^\infty x e^{-x^2} dx$$

$$= 1/2 \int_0^\infty e^{-t} dt$$

$$= 1/2 \left(\frac{e^{-t}}{-1} \right)_0^\infty$$

$$= -1/2 (e^{-\infty} - e^0)$$

$$= 1/2$$